

Low-Power Hyperbolic CORDIC Design of the Fitzhugh-Nagumo Neuron for Neuromorphic Systems

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Abstract—Neuromorphic hardware designs require hardware-efficient and biologically inspired neuron models to achieve accurate and low-power computation. However, existing implementations of the FitzHugh–Nagumo (FHN) neuron often encounter hyperbolic CORDIC resource utilization, energy efficiency, and accuracy in reproducing nonlinear dynamics. This paper presents a novel low-power FHN neuron design based on a hyperbolic CORDIC algorithm, which approximates the nonlinear term using a simplified polynomial-based hyperbolic function. The design employs fixed-point arithmetic and shift-and-add operations for efficient FPGA implementation. The proposed neuron achieves high accuracy, with an NRMSE of 0.1788, while significantly reducing hardware resources to 138 LUTs and 116 FFs, lowering power consumption to 12.05 mW, and operating at 398 MHz with 22.6 ns latency. These results demonstrate that the proposed hyperbolic CORDIC-based FHN neuron provides a high-performance and accurate solution for low-power neuromorphic hardware systems.

Index Terms—Spiking Neural Networks, FitzHugh–Nagumo neuron, multiplier-free design, hyperbolic CORDIC, low-power FPGA, neuromorphic computing.

I. INTRODUCTION

Spiking Neural Networks (SNNs) have attracted considerable research interest due to their potential for energy-efficient neuromorphic hardware implementations [1]. Spiking neurons are commonly represented through differential equations. The Hodgkin–Huxley (HH) model provides a highly biologically accurate description of neuronal activity, albeit at a substantially higher computational cost [2]. Compared with the HH model, the FitzHugh–Nagumo (FHN) model offers reduced complexity while preserving excitability and spike generation [3], [4]. Because of its efficiency and biological relevance, the FHN model is widely used to study neuronal spiking and bursting dynamics [5], as well as in neuromorphic circuit design for low-power computing applications [6]. Recent research has explored efficient digital realizations of the FHN model, each aiming to improve hardware efficiency and computational accuracy through different techniques [7]–[13]. Early works introduced multiplierless designs to implement the FHN neuron, reducing hardware resource usage [7]–[10]. Subsequently, CORDIC-based implementations were introduced to approximate the nonlinear terms of the FHN model [11], [12]. These implementations rely on the CORDIC

algorithm, an iterative shift-and-add technique used to efficiently compute nonlinear functions in digital hardware [14]. Additionally, floating-point implementations, were developed to improve computational precision [8]. More recently, low-power FPGA implementations have leveraged power-of-two arithmetic to minimize energy consumption [9]. Despite these advances, existing digital FHN realizations still face trade-offs between accuracy, hardware cost, and power efficiency. Therefore, we propose a novel FHN neuron implementation based on a new hyperbolic CORDIC algorithm, offering a multiplier-free design that balances computational accuracy, resource utilization, and power consumption to tackle these limitations.

Unlike prior FPGA-based FHN implementations relying on multipliers, LUTs, or polynomial approximations, the proposed method reformulates the cubic nonlinear term into a $\sinh$ -based expression compatible with Hyperbolic CORDIC, enabling a multiplier-free, pipeline-friendly architecture with reduced hardware cost and power while maintaining competitive accuracy.

The following sections outline the structure of the paper. Section II provides a review of the original FHN model and the CORDIC algorithm. Section III details the proposed model, while Section IV analyzes its dynamic characteristics and evaluates the associated error. Section V describes the hardware implementation and presents the corresponding results. Finally, Section VI summarizes the findings and concludes the paper.

II. BACKGROUND

A. The Fitzhugh-Nagumo Neuron Model

The dynamics of the FHN neuron are described by the following differential equations:

$$\begin{cases} \frac{dv}{dt} = f(v) - w + I \\ \tau \frac{dw}{dt} = v + a - bw \\ f(v) = v - \frac{v^3}{3} \end{cases} \quad (1)$$

Here, v represents the membrane potential of the neuron, and w is a recovery variable that models slower ionic processes. The function $f(v)$ introduces the nonlinear spiking behavior, I is the input stimulus current, τ is a time constant, and a and b are dimensionless parameters controlling the shape and threshold of neuronal excitability.

B. The CORDIC Algorithm

The iterative equations for hyperbolic CORDIC are:

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Procedure proposed hyperbolic CORDIC to calculate $4\sinh(0.75v)$

Input:
 $z_{in} : 0.75v, v \in \{-2, 2\}$ //Input variable to the hyperbolic sinh function
CORDIC angles lookup table LUT $//LUT[i] = \tanh^{-1}(2^{-i})$

Output:
 $y_{out} : 4\sinh(0.75v)$

1. $K^{-1} = 1.2051$ // K is the scale factor
2. **if** $|z_{in}| \leq 1.1182$ **then** // Initialization
3. $y \leftarrow 0; x \leftarrow 4K^{-1}; z \leftarrow |z_{in}|;$
4. **else**
5. $y \leftarrow 4K^{-1}\sinh(1.1182); x \leftarrow 4K^{-1}\cosh(1.1182); z \leftarrow |z_{in}| - 1.1182$
6. **for** $i = 1:n$ **do**
7. **if** $z > 0$ **then**
8. $y = y + x \cdot 2^{-i}; x = x + y \cdot 2^{-i}; z = z - LUT[i];$ // In parallel
9. **else**
10. $y = y - x \cdot 2^{-i}; x = x - y \cdot 2^{-i}; z = z + LUT[i];$ // In parallel
11. **end if;**
12. **end for;**
13. **if** $z_{in} \geq 0$ **then**
14. **return** $y;$ // $y_{out} = y$
15. **else**
16. **return** $-y;$ // $y_{out} = -y$

Fig. 1. Pseudocode of the hyperbolic CORDIC algorithm for computing $4\sinh(0.75v)$

$$\begin{cases} x_{i+1} = x_i + d_i y_i 2^{-i} \\ y_{i+1} = y_i + d_i x_i 2^{-i} \\ z_{i+1} = z_i - d_i \tanh^{-1}(2^{-i}) \end{cases} \quad (2)$$

where x_i and y_i are the components of a vector being rotated in the hyperbolic plane, z_i denotes the remaining angle, and $d_i = \pm 1$ is the rotation direction selected according to the sign of the residual angle z_i to progressively reduce it toward zero. The factor 2^{-i} corresponds to a binary-scaled factor, enabling each iteration to be implemented using simple shift-and-add operations. After n iterations, when z_n becomes negligible, the outputs x_n and y_n are approximately expressed as [14]:

$$\begin{pmatrix} x_n \\ y_n \end{pmatrix} = K \begin{pmatrix} \cosh(z_0) & \sinh(z_0) \\ \sinh(z_0) & \cosh(z_0) \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \quad (3)$$

where K is the CORDIC scale factor arising from the slight change in vector magnitude during the iterative rotations [14]. To eliminate the scale factor K , the initial values x_0 and y_0 are pre-scaled by K^{-1} , i.e., $x_0 = K^{-1}$ and $y_0 = 0$, while z_0 is set to the input z_{in} , which lies within the standard hyperbolic CORDIC range $(-1.1182, 1.1182)$ [14]. Therefore, with this initialization, the outputs become [14]:

$$x_n = \cosh(z_{in}) \text{ and } y_n = \sinh(z_{in}) \quad [14]$$

III. PROPOSED MODEL

This section presents an improved version of the FHN neuron model, in which the original nonlinear function $f(v)$ in (1) is substituted with a simplified function $g(v)$. Since the cubic term in $f(v)$ closely resembles the behavior of a hyperbolic sine function, the proposed approximation is formulated as:

$$g(v) = c_1 v - c_1 \sinh(c_2 v) \quad (4)$$

where c_1 and c_2 are coefficients and v denotes the neuron's membrane potential. The hyperbolic sine function can be approximated using its Taylor series, and to simplify the approximation while retaining the main nonlinear behavior of $f(v)$, only the first two terms are considered and substituted into $g(v)$:

$$g(v) \approx c_1(1 - c_2)v - \frac{c_1(c_2 v)^3}{3!} \quad (5)$$

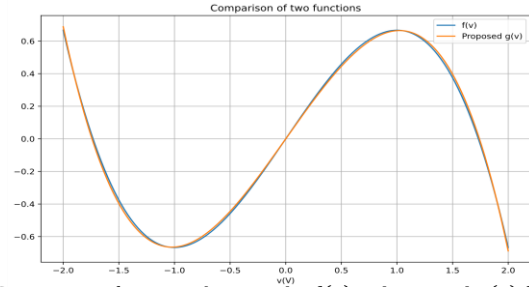


Fig. 2. Comparison of accuracy between the $f(v)$ and proposed $g(v)$ functions.

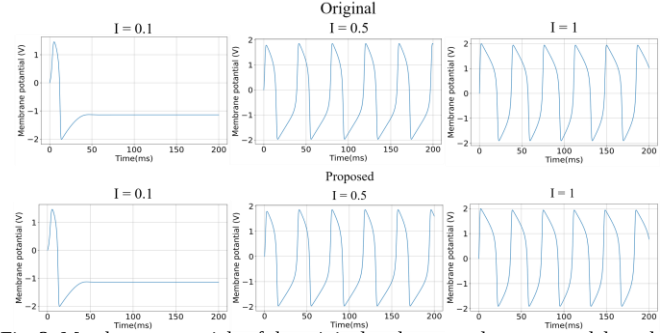


Fig. 3. Membrane potentials of the original and proposed neuron models under varying input currents.

The coefficients c_1 and c_2 are determined by matching the linear and cubic terms of $g(v)$ with those of $f(v)$. Solving the resulting equations yields $c_1 = 4.36$ and $c_2 = 0.775$. To reduce computational complexity and hardware resource utilization, these values are rounded to $c_1 = 4$ and $c_2 = 0.75$. Thus, the proposed function is expressed as:

$$g(v) = 4v - 4 \sinh(0.75v) \quad (6)$$

The nonlinear term $4 \sinh(0.75v)$ in $g(v)$ is computed using the proposed hyperbolic CORDIC algorithm, as shown in Fig. 1. Since the input range $z_{in} = 0.75v, v \in (-2, 2)$, exceeds the hyperbolic CORDIC's standard range, we divide the input into a couple of intervals and assign different hyperbolic initial values to each, following the approach in [15]. Due to the odd property of the $\sinh$ function, $\sinh(-x) = -\sinh(x)$, the CORDIC algorithm operates only on the input magnitude. After convergence, the output $y = 4 \sinh(0.75v) = 4 \sinh(m + n)$ is expressed as:

$$y = K(\sinh(m) \cosh(n)) \begin{pmatrix} 4K^{-1} \cosh(n) \\ 4K^{-1} \sinh(n) \end{pmatrix} \quad (7)$$

where m and n are obtained by decomposing the input $z_{in} = 0.75v$, such that $z_{in} = m + n$, and the inverse of the scale factor is $K^{-1} = 1.2051$. Equation (7) generalizes the hyperbolic CORDIC relation in (3). This implies that, even when the input angle exceeds the predefined hyperbolic CORDIC range, the CORDIC algorithm can still be employed by appropriately adjusting the initial conditions. After initialization, the values of x , y , and z are updated through iterative rotations in the CORDIC algorithm, yielding the final output $y = 4 \sinh(0.75v)$. This approach enables computation of the hyperbolic functions $\sinh$ and $\cosh$ over an extended input range of $(-2.2364, 2.2364)$. As shown in Fig. , the proposed function $g(v)$ reveals close alignment with the original function $f(v)$ across the voltage range of $(-2, 2)$, with

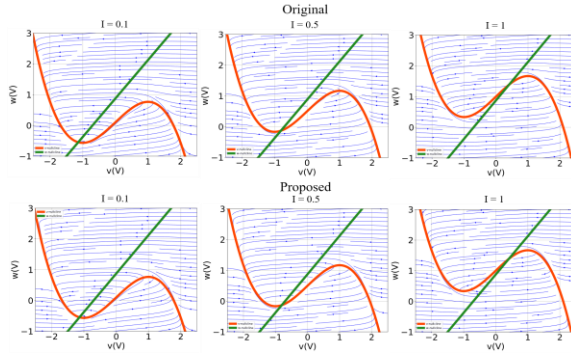


Fig. 4. Original vs. proposed model phase portraits.

a root-mean-square error (RMSE) of 0.012, and Fig. confirms that the spiking activity of the proposed model closely resembles that of the original FHN neuron, thereby validating both the accuracy and efficiency of the approximation.

IV. MODEL EVALUATION

Software simulations are employed in this section to validate the proposed model. The evaluation covers error analysis, as well as the model's dynamic behavior.

A. Error Analysis

In this section, error analysis, including RMSE, Mean Absolute Error (MAE), and Normalized Root Mean Square Error (NRMSE), is performed to evaluate the proposed model against the original FHN model. Table I summarizes RMSE, MAE, and NRMSE values under various input conditions ($a = 0.7, b = 0.8, \tau = 10$). The results confirm that the proposed model accurately reproduces the dynamic behavior of the original FHN model.

Moreover, prior network-level studies of approximate FHN neurons show that moderate neuron-level errors are generally absorbed by the inherent robustness of distributed spiking dynamics [8], [9]. Hence, the achieved NRMSE reflects a trade-off between accuracy and hardware efficiency.

Additional analyses indicate that the dominant error arises from the sinh-based approximation, while other sources have minor impact. Owing to the model's nonlinear recursive dynamics, these errors are not linearly additive, resulting in a final NRMSE of 0.1788.

B. Dynamic Behavior

The dynamic analysis of both the proposed and original models is performed through equilibrium points and the phase portrait, the latter illustrating the system's qualitative behavior, including trajectories, while the former—which are also visible in the phase portrait—are computed by setting $dv/dt = 0$ and $dw/dt = 0$; the original model's nullclines are provided in [12], whereas those of the proposed model are presented as follows:

$$\begin{cases} \frac{dv}{dt} = 0 \Rightarrow g(v) + I = w \\ \frac{dw}{dt} = 0 \Rightarrow \frac{v + a}{b} = w \end{cases} \quad (8)$$

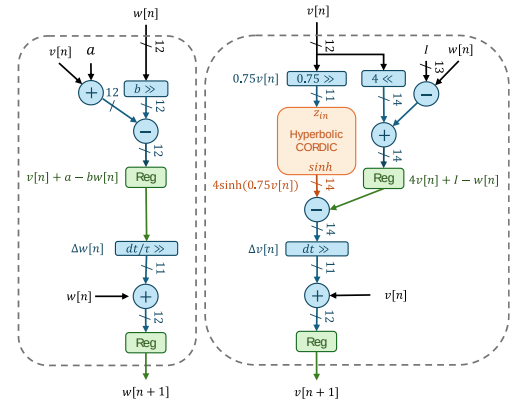


Fig. 5. Proposed multiplier-free CORDIC-based FHN neuron model architecture.

Table I
ERROR VALUES OF THE PROPOSED MODEL

I(mA)	MAE	RMSE	NRMSE %
0.1	0.005	0.011	0.249
0.25	0.014	0.016	0.209
0.5	0.020	0.021	0.110
0.75	0.024	0.031	0.163
1	0.029	0.031	0.163
Average	0.018	0.022	0.1788

Table II
EQUILIBRIUM POINTS COMPARISON

I(mA)	Original (v, w)	Proposed (v, w)
0.1	Spiral sink (-1,1375, -0.5469)	Spiral sink (-1,1374, -0.5468)
0.5	Spiral source (-0.8048, -0.1311)	Spiral source (-0.8046, -0.1310)
1	Nodal source (0.4089, 1.3861)	Nodal source (0.4088, 1.3860)

As shown in Table II, the proposed model accurately reproduces the equilibrium points and their numerical values of the original FHN model, with these points varying according to changes in the bifurcation parameter. Fig. 4 also shows that the proposed model correctly replicates the dynamic behavior of the original system under applied currents of 0.1, 0.5, and 1 mA.

V. HARDWARE IMPLEMENTATION

This section presents the FPGA implementation of the proposed neuron model, highlighting the digital design process using Euler discretization and the corresponding architecture.

A. Neuron Architecture

The proposed neuron dynamics are discretized using the Euler method as follows:

$$V[n + 1] = V[n] + dt(4V[n] + I[n] - W[n] - \text{CORDIC}(0.75V[n])) \quad (9)$$

$$W[n + 1] = W[n] + dt(V[n] + a - bW[n])/ \tau \quad (10)$$

where the algorithm of $\text{CORDIC}(0.75V[n])$ in (9), used to calculate $4 \sinh(0.75V[n])$, is shown in Fig. 1. In digital implementation, fixed-point arithmetic is employed to ensure efficient hardware realization and the design relies solely on addition, subtraction, and logical shift operations, minimizing hardware cost compared to multiplier-based designs, as illustrated in Fig. . For instance, the $0.75 \gg$ and $4 \ll$ blocks, representing multiplication operations in (9), are implemented

Table III

HARDWARE IMPLEMENTATION RESULTS FOR FHN NEURON MODELS

Model	LUTs	FFs	Fmax (MHz)	Power (mW)	Latency (ns)	NRMSE (%)
[7] (Virtex-2)	1085	526	-	-	-	-
[8] (Zynq)	375	221	450	-	13.3 (R)	0.00483
[9] (Zynq)	140	141	224.56	26	13.36 (R)	0.36
[10] (Virtex-2)	912	457	175	-	-	1.25
[11] (Kintex)	306	226	367.78	-	87 (E)	0.1404
[12] (Virtex-4)	161	216	320	295.49	37.5 (E)	1.43
[13] (Virtex-2)	543	664	-	-	-	3.42
This work (Zynq)	138	116	398	12.05	22.6 (R)	0.1788

Note: Latency values marked as (R) are reported, whereas (E) indicates estimated values.

using shift operations. Moreover, the time step dt is set to 2^{-7} and the time constant τ is selected as 2. As a result, the expression dt/τ in Fig. is implemented as a single 8-bit right shift, simplifying the architecture.

To optimize numerical precision and resource efficiency, this architecture employs fixed-point arithmetic with a 9-bit fractional part, supporting up to 8-bit right-shift operations without data loss. The integer bit-width is selected according to the dynamic range of each signal. When the dynamic range of a signal is small, allocating many bits for the integer part would be unnecessary, as most would remain unused. To maintain both accuracy and hardware efficiency, the integer bit-width is chosen to be just large enough to represent the maximum magnitude of the signal. This ensures that no resources are wasted on unused bits while still preventing overflow. The CORDIC function in (9) is implemented using a CORDIC unit to compute $4 \sinh(0.75V[n])$ through iterative shift-and-add operations. In the hyperbolic CORDIC operation, we adopt an 8-bit precision. Since the architecture integrates a CORDIC unit, data gating is applied to ensure reliable computations. To achieve a low-latency design, computations are parallelized, and registers are used to pipeline the structure, mitigating the critical path delay and enhancing the maximum operating frequency, as shown in Fig. .

B. Results

This section presents the hardware implementation results of the proposed FHN neuron model on the Xilinx Zynq xc7z100 FPGA. Table III compares the results of the proposed model with previous FHN implementations. Compared with the latest work [9], the proposed FHN neuron achieves slightly lower LUT usage (138 vs. 140) while reducing FFs by 17.7%. It also operates at a frequency of 398 MHz, 77.3% faster than [9], while consuming 12.05 mW, a 53.7% reduction, and achieving a 50.3% improvement in accuracy. Although the latency, calculated using the same methodology as in [9], is inherently increased due to CORDIC iterations, the proposed design still achieves 22.6 ns, which is lower than the latency of other CORDIC-based implementations, 87 ns [11] and 37.5 ns [12]. For a typical neuromorphic time step of $1 \mu s$, the proposed architecture allows sequential time-multiplexed sharing of a single CORDIC unit among approximately 44 neurons, indicating that the reported latency does not impose a throughput bottleneck. Overall, the results demonstrate that the proposed design outperforms all prior works [7]–[13]. It should be noted that the latency values for these prior works were

estimated in [9], as they were not originally reported, and entries marked with a hyphen (-) in Table III indicate metrics that were not provided in the cited works. The proposed design adopts an algorithm–hardware co-design approach based on a Hyperbolic CORDIC-compatible reformulation of the nonlinear term, reducing hardware complexity and power consumption while maintaining competitive accuracy.

VI. CONCLUSION

In conclusion, we proposed an optimized hardware implementation of the FHN neuron model tailored for low-power and high-accuracy applications. By replacing the nonlinear term with a simplified polynomial approximation and computing it using hyperbolic CORDIC architecture, the design achieves reduced power consumption and improved computational accuracy.

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